

K. PRAJITH RAJ

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PROFESSIONAL SUMMARY

ECE undergraduate with hands-on **experience in wireless communication, embedded systems, and IoT**. Skilled in **LoRa, ESP32, Raspberry Pi, MQTT, and communication protocols**, with a strong foundation in **RF fundamentals, signal transmission, and electronics**. Experienced in developing and testing long-range wireless communication systems through projects. Seeking an RF Engineer role to contribute to communication infrastructure.

EDUCATION

St. Joseph's Institute of Technology, Chennai
Bachelor of Engineering in Electronics & Communication Engineering

2023 – 2027
CGPA: 7.8 / 10

EXPERIENCE

Incax Technology Pvt. Ltd., Chennai – *Embedded Systems & IoT Intern*

Oct 2025 – Apr 2026

- Engineered a full-stack IoT Smart EV Charging Station — integrating ESP32 microcontroller with cloud backend for automated charging sessions, user authentication, and energy consumption tracking, delivering a functional product prototype.
- Designed responsive UI and optimised energy-management firmware algorithms, reducing average charge-cycle time by ~15% and ensuring stable performance for 50+ concurrent users.

HPRCSELab, IITDM Kancheepuram – *Embedded Systems Intern*

Jun 2025- Jul 2025

- Developed an IoT Digital Power Meter using ESP32 that measured voltage, current, power, and energy across 3 load types in real time with $\pm 2\%$ measurement accuracy, deployed with a live cloud monitoring dashboard.
- Integrated threshold-based overvoltage/undervoltage fault detection triggering automatic relay cutoff within 200 ms — reducing manual inspection effort by ~60% and preventing equipment damage.

PROJECTS

LoRa-Based Wireless Communication for Coal Miners | *ESP32, LoRa SX1276, Embedded C*

2025

- Built a sub-GHz LoRa mesh network enabling bidirectional data transmission over 2km in GPS/Wi-Fi-denied underground mine environments — completely eliminating communication blackouts for miners inside tunnels.
- Achieved ~70% reduction in communication blackout zones during tunnel simulation tests; enabled real-time miner location tracking, directly improving emergency response capability in high-risk environments.

Parking Management Web App – Thoothukudi Police Hackathon | *Raspberry Pi, OpenCV, React, MongoDB*

2024

- Built an end-to-end smart parking solution: Raspberry Pi + OpenCV ML camera achieving 92% Car Parking area, React frontend, MongoDB backend, and a pre-booking engine that processed 200+ slot reservations, reducing urban traffic congestion by ~30%.

IoT-Based Smart Cow Identification & Tracking System | *ESP32, IoT, GPS, RFID Tag, Wireless Communication*

2025

- Developed an IoT-enabled smart cattle monitoring system using electronic tags and ESP32, enabling unique identification and real-time tracking of individual cattle through wireless data transmission.
- Designed a low-power, scalable communication architecture for continuous livestock monitoring and digital record management, reducing dependency on manual cattle identification and tracking.

SKILLS

Core Electronics: Analog Circuits, QNX, Transistors, ADC/DAC, Antenna, Signal Conditioning

MCUs & SBCs: ESP32, Arduino Uno/Nano, Raspberry Pi, STM32, Embedded Linux, Embedded C, real-time control, sensors

Communication: I²C, UART, SPI, MQTT

Design & Simulation: MATLAB, Simulink, Proteus, Easy EDA, ISM, Keil

Programming: C, C++, Python | Full Stack: React, Node.js, MongoDB

CERTIFICATIONS

- Embedded Systems & IoT – edX
- Microcontrollers & Microprocessors – Udemy
- Python For Data Science-NPTEL
- HTML, CSS & JavaScript – Infosys Springboard
- Database Management Systems – NPTEL
- Embedded System-Microchip

ACHIEVEMENTS & AWARDS

- 1st Prize – CAG Ideathon (12-hour national hackathon)
- 2nd Place – Becrez'25 National Hackathon (₹5,000 prize + ₹24,000 credits)
- 26th Rank – IIT Bombay International Hackathon 2026
- 2nd Place – Shark Tank Style Business Idea Competition
- Won 3+ National-Level Technical Symposium Events
- Asia Level Entrepreneurship Hackathon Shortlisted -IIT Bombay